

Management

FORM FOR THE SUBMISSION OF ASSESSED GROUP COURSEWORK

THIS FORM MUST BE COMPLETED AND ATTACHED TO **ALL** ASSESSED
WORK.

Please include all group members' student numbers below.

Student Number	2	4	0	3	2	9	6
Student Number	2	3	8	2	6	1	2
Student Number	2	3	9	3	0	8	1
Student Number	2	3	5	7	2	6	5

**We agree / do not agree* that equal and fair contributions were made to the
final report by all members of our group.**

* Please delete as appropriate. In the event that all members of the group do not agree that they have worked together effectively, shared the workload equitably and are therefore happy to receive the same individual mark for the project, they will be required to provide written evidence as to why this occurred.

Please list brief details of how tasks were divided up among the group members in the box below. (Please refer to chapters of final report etc.)

By submitting this work online using my unique log-in and password, I confirm that I am submitting the work on behalf of my group and that it is all our own work.

We understand that all marks are provisional until ratified by the Faculty Examination Board.

Please remember to save **your work** either as a **Word document (.doc)** or as a **PDF (.pdf)** unless it is a presentation or recording where it should be saved in an appropriate format for the work with the filename [Unit Code Group Number or Name] e.g. **MGRCM0000_Group 1.docx**.

Please start writing your coursework on the next page.

Team Fundraising Catalyst : Understanding the Fundraising campaigns through an Analytical approach

Analytics Problem

Problem Identification:

In the face of economic volatility and changing donor behaviours charities are facing the problem of evaluating and improving the efficiency of their fundraising campaigns. Due to the global recession, regular benefactors are facing an economic breakdown due to which charities are facing a decline in donations and grants hence people dependent on them are affected adversely (Preston, 2024). With the evolve of modern methodologies, conventional methods may not be enough so it's important to methodically identify the factors that influence successful fundraising, properly customise donor communications, and make sure that resources are used as efficiently as possible. The core problem we focus on is how to apply machine learning and other modern-day technologies to better comprehend and enhance the effectiveness of fundraising campaigns and automate the process.

Introduction:

According to IBIS World, a leading global industry research firm, the UK charity sector is valued over £50 billion and continues to grow in 2023. However, behind the growth lie significant challenges that include issues like limited resources, outdated donor communication strategies, poor marketing and a lack of tailored communication strategies resulting in loss of opportunities and reduction of investment returns (Quinton et al., 2013). With decreasing donations and increasing demand for charitable services, place charitable organizations in a critical place which offers an opportunity for innovative technology to enhance the effectiveness of fundraising campaigns through data-driven solutions to understand donor behaviour, analysing factors impacting success of charity campaigns and utilize these insights to devise strategies for efficient donor engagement (Bhati and McDonnell, 2019).

In this project we have used logistic and linear regression to establish a basic relationship between features useful for initial insights, ensemble methods like random forest and gradient boosting were used for predicting donor behaviour and optimising grant allocation, random forest and support vector machine(SVM) were used to build initial models to accurately analyse

and predict outcomes. Donor analysis on demographic data from government census plays a crucial role by enabling the identification and prediction of potential donors enhancing the efficiency of outreach efforts and sentiment analysis on twitter data offers valuable insights into public perception and helps the organisation through adjustments in current communication strategies. Finally, we tried to set up an automated tableau dashboard by combining the TabPy library, which can create a link between python and tableau. Any changes to the code file will be reflected dynamically in tableau. This dashboard will include descriptive analysis such as distribution of different fundraising themes and public's sentiment toward different social issues. This dashboard enables charities to evaluate the effectiveness of fundraising campaigns in a real-time and automated approach will help them reduce manual work. By applying these techniques we not only expect to improve donor engagement but also improve the efficiency of fundraising campaigns, guaranteeing those fundraising organizations continuously fulfil their vital role to the society. For this project, Duro Insight a start-up that works in the domain of reputation insight and competitor analysis has partnered with the University of Bristol Business School, representing the strategic initiative to integrate advanced data analysis into the charity domain.

Technical Development

In order to help charities better comprehend and enhance the effectiveness of fundraising campaigns, it is crucial to identify the demographic characteristics of donors who are more likely to make donations (Maftai, 2020). We did an exploratory data analysis (EDA) and initial machine learning (ML) with Random Forest Classifier and KNN on the dataset that we collected from GrantNav (360 Giving). The EDA involves cleaning the data, identifying distributions of grants over time, regions, sectors, and weekdays, and examining the relationship between different variables using density plots, pie charts and bar charts.

Further we used the google Prophet model to do predictive analysis using time series forecast on the historical data and get prediction for the donation activity in the coming years. By utilizing demographic information from the charities we can apply targeted marketing strategies to reduce marketing expenses and enhance effectiveness of fundraising initiatives. In order to accomplish this objective we used Logistic Regression and Random Forest Classifier, XGB Classifier and MLP Classifier to predict potential donors. We got information regarding

donor location, education, gender and other features to help charities identify those most suitable potential donors.

Once potential donors have been identified, the next aspect that should be considered is how to allocate resources dynamically to organize fundraising campaigns. The combined approaches of sentiment analysis and word cloud technology were able to meet this requirement. Sentiment analysis is a type of natural language processing that is used to identify emotions in the text descriptions (Kumawat, Yadav, Pahal, & Goel, 2021). To accomplish this, we utilized TextBlob to analyze the sentiment of the text. By creating a TextBlob object and accessing its sentiment attribute, we obtained the polarity value, which indicates whether the sentiment is positive, negative, or neutral. As for word cloud technology, we used the Word Cloud library in python to arrange keywords in text by frequency to identify those fundraising themes that attract the public's attention. By combining this with sentiment analysis, we are able to compare the sentiment distributions across fundraising themes to understand the public's attitudes towards different themes and allocate their resources accordingly.

Considering that the distribution of donors and public sentiment toward fundraising themes fluctuate over time, it is necessary to dynamically visualize the demographic characteristics of donors and sentiment distribution of fundraising campaigns. As a result, Tableau dashboard was applied. It allows for flexible time segmentation (Balaji,2021) and offers visual insights into donor information and the sentiment landscape of fundraising campaigns over specific time frames. Additionally, by applying the TabPy library of Python, we successfully created a link between Python and Tableau that any changes to the dataset and code file in Python will be reflected on the dashboard of Tableau, which will significantly save time from repetitively analysing same dataset.

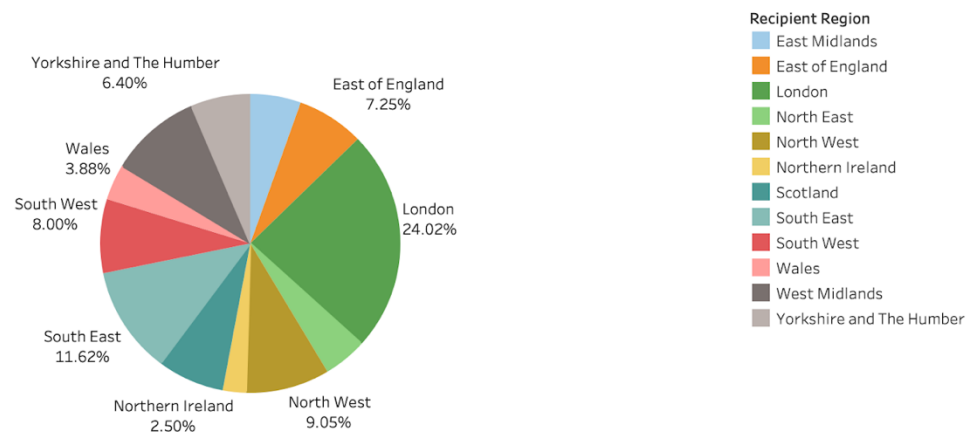
In addition to this in the present landscape, it is of equal significance to forecast the future scope of fundraising campaigns and propose strategies accordingly. As a result, we leveraged the Prophet library to predict donation over the years of fundraising campaigns. By doing this, charities are able to accurately estimate future donor demographics and sentiment dynamics across various categories of charitable activities, facilitating more targeted resource allocation.

Analytical Solution:

DEMOGRAPHIC AND DONOR ANALYSIS:

Analysis on the basis of Region wise Funding:

1. By doing region wise funding analysis we found that London has received the maximum number grants which accounts for 24.02% of total grants.
2. Followed by South East, West Midlands and Northwest at 11.62, 9.92 % and 9.05% respectively.
3. While Wales has received 3.88 % of grants which is the least number of grants.



Recipient Region and % of Total Count of Recipient Region. Color shows details about Recipient Region. Size shows count of Recipient Region. The marks are labeled by Recipient Region and % of Total Count of Recipient Region. The view is filtered on Recipient Region, which excludes Null and England.

Figure-1: The above graph represents the percentage of funding received from various regions across the UK.

Analysis on the basis of Month wise Donation Activity:

1. By doing Month wise Donation analysis we found that April has the maximum number of donor activity which accounts for 32.11% of total grants followed by June and November at 8.19% and 7.43% respectively.
2. We also found that donations in the month of August is the least at 4.33%.

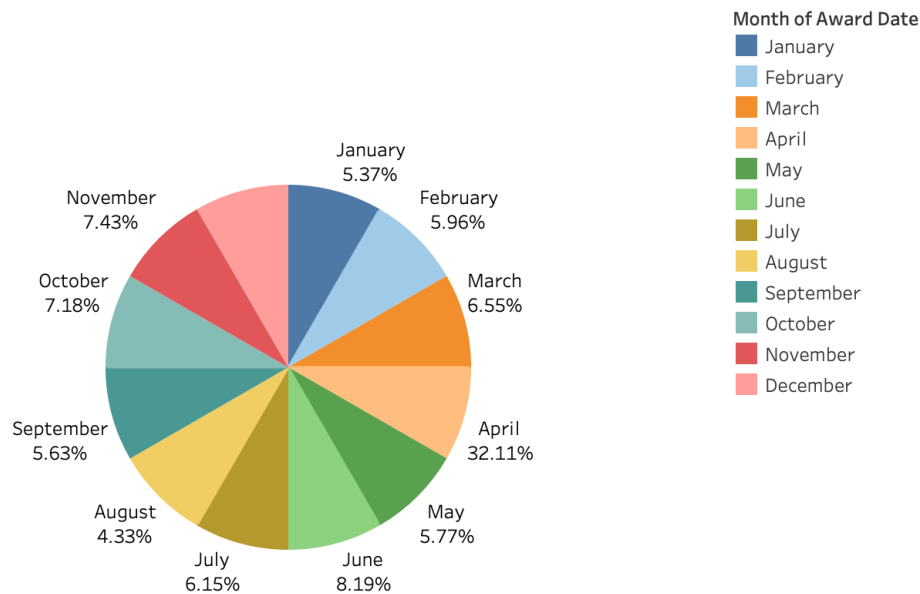


Figure-2: The above graph represents the percentage of funding received across various months.

Analysis on the basis of Income vs Age groups of individuals:

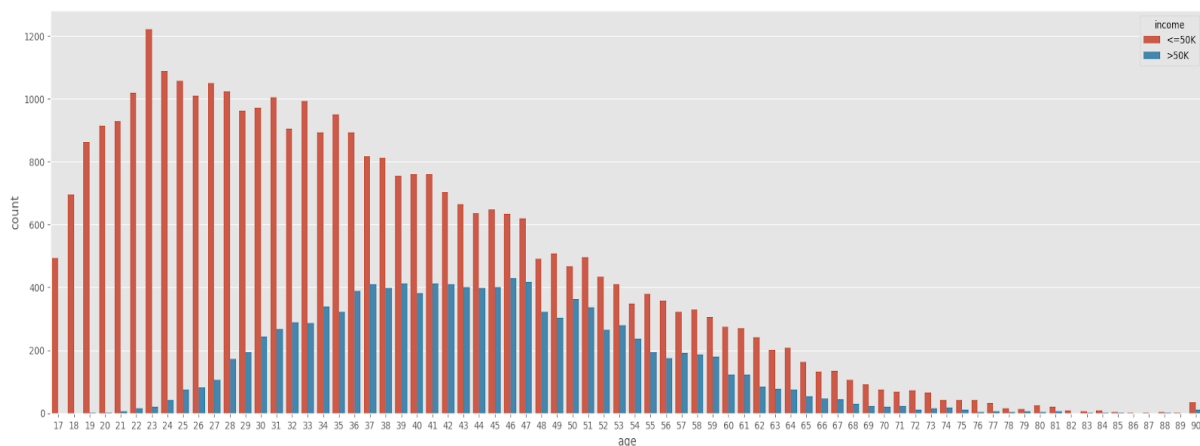


Figure-3: Graph showing relation between Income and age of individuals.

1. The plot above shows the relationship between salaries and age of individuals.
2. Maximum people in the age group of 37 and 47 belong to the higher earning category earning more than 50 thousand pounds per annum and are expected to be potential donors.
3. It shows that maximum individuals at the age 23 years old belong to the lower income category earning less than 50 thousand pounds per annum.

- Thus, individuals between the ages of 35 and 47 stand at a higher position to contribute and thus should be targeted.

Analysis on the basis of gender:

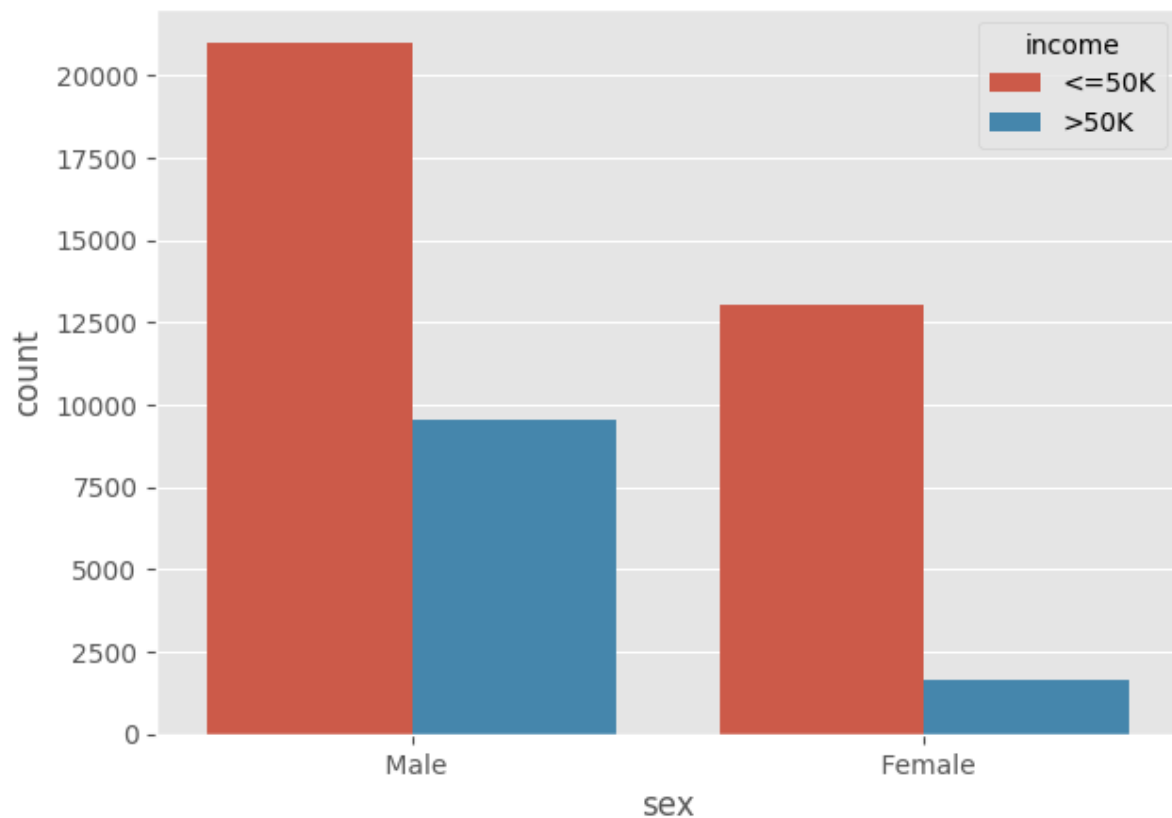


Figure-4: The above graph represents gender wise Income differences.

- The above graph is comparing individuals on the basis of gender and income levels
- There are more males individuals than females individuals in both income categories.
- For both the gender categories the count of individuals earning 50 thousand or less is higher than those earning more than 50 thousand pounds.
- The number of males earning greater than 50 thousand pounds per annum is larger than the females.

Analysis on the basis of education levels:

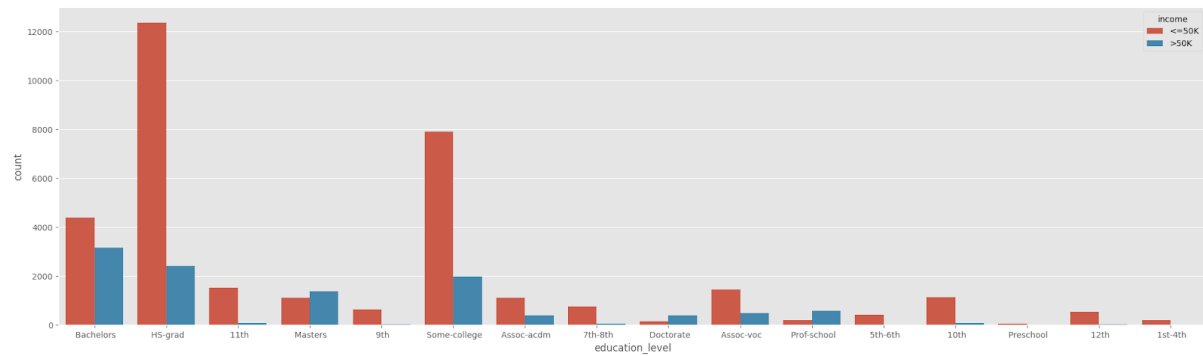


Figure-5: Visualization showing Income across the Education levels.

1. The above graph represents income across various income levels.
2. The visualization shows that the maximum number of individuals with bachelor degree earn more than 50 thousand pounds per annum.
3. It also shows that maximum number individuals with High School Degrees earn less than 50 thousand pounds per annum followed by individuals with some college education and individuals with just bachelor degree.

Growth of Charity activity over the years (Historical vs Predictive):

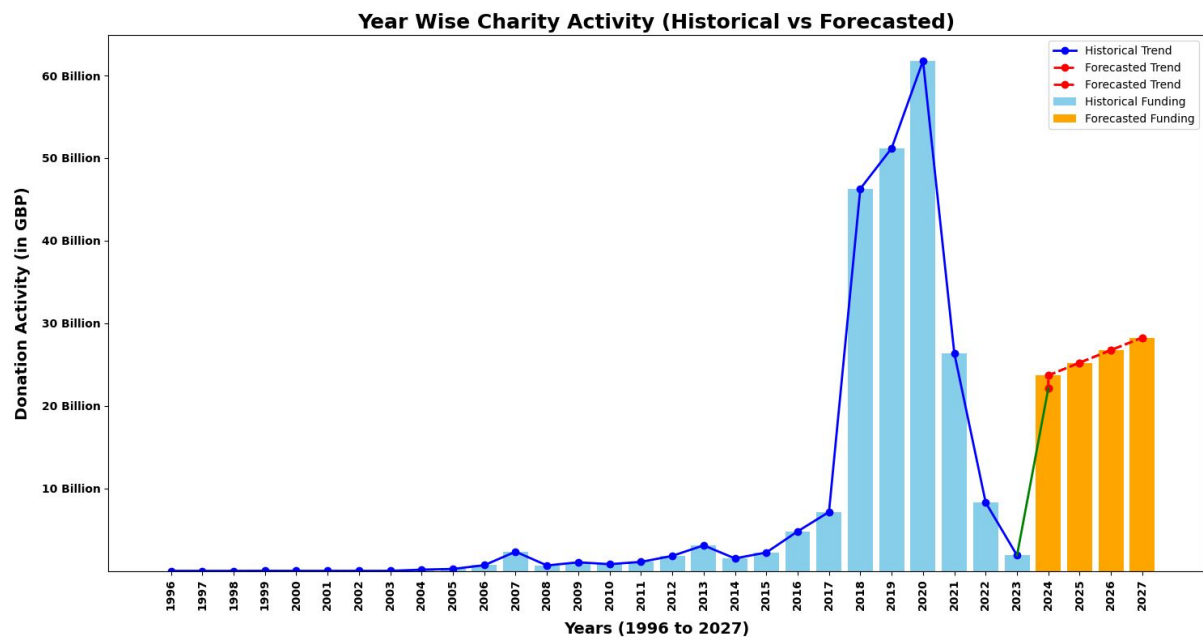


Figure-6: Historical vs Predicted Year wise Charity Activity

1. The analysis shows that charity activities across the last three decades, from the year 2006 to present.
2. The blue bars represent historical data, showing the fluctuations in donations received across the last thirty years and the orange graph shows the forecasting of donations in the next few years.
3. By using the Google Prophet model we did predictive analysis and we forecasted that we are going to see a steady recovery in donation activity in the coming years from 2024 to 2027.

Analysis on the basis of employment sector and income:

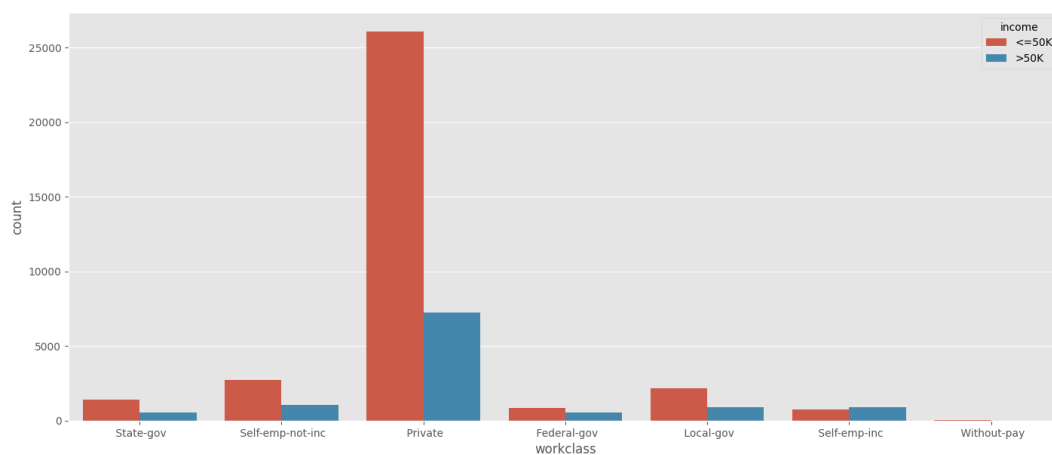


Figure-7: Above graph represents analysis on the basis of Employment Sector and income.

1. From the above it can be found that people working in the private sector have higher income and the lowest being the individuals with no pay.
2. In the private sector over 25000 individuals have an income less than equal to 50,000 pounds per annum and around 7500 individuals have an income greater than 50,000 pounds.
3. The private sector still remains the highest in terms of number individuals earning more than 50 thousand pounds per annum.

Analysis on the basis of number of donations given:

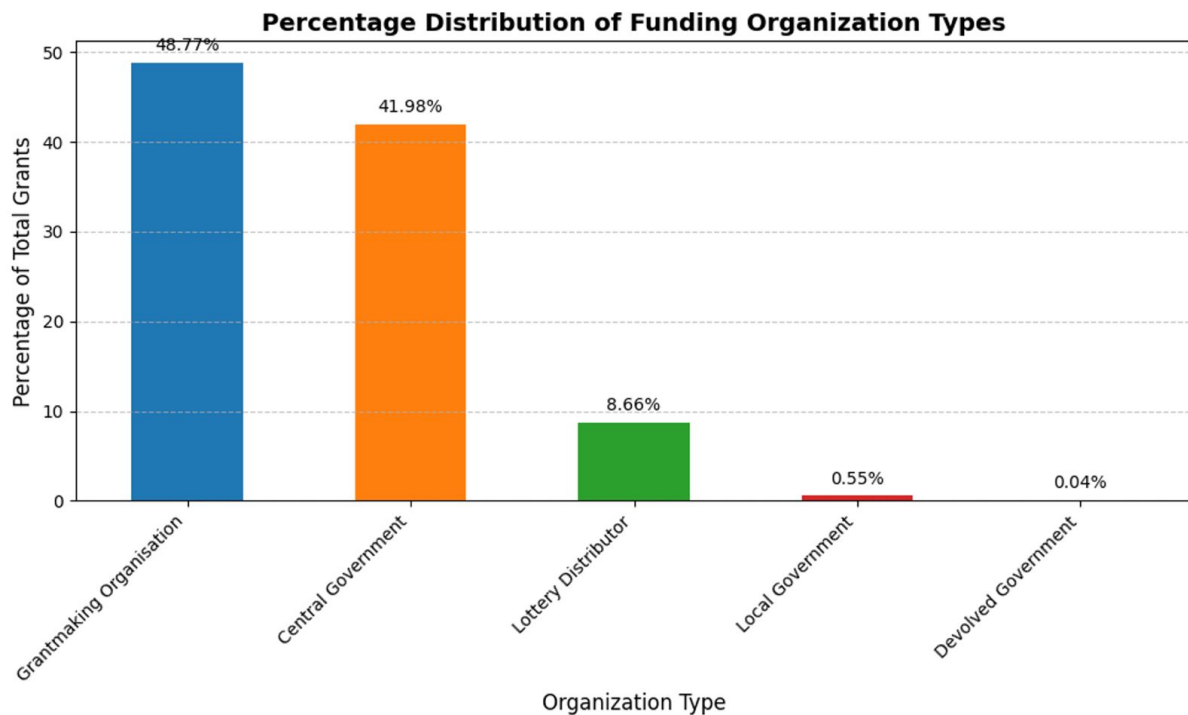


Figure-8: Percentage Distribution of Funding Organizations on the basis of number of donations given.

1. Based on the percentage of grants among funding organization types , we have five categories Grant making organisation, central government, Lottery distributor and devolved governments.
2. Grant making Organizations receive maximum donations accounting for 48.77% of total grants while devolved government bodies receive the least number of grants around 0.04 %.

SENTIMENT ANALYSIS

Analysis based on the type of sentiments :

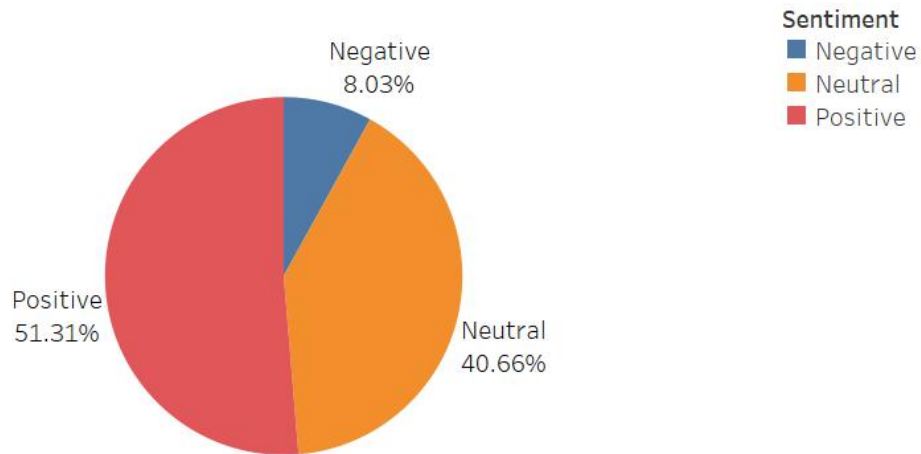


Figure-9: Figure showing the distribution of sentiments

1. The above pie chart indicates that 51.3% of sentiments are positive, while 40.6% are neutral and only a small number of them around 8.03% are negative sentiments.
2. Thus it can be observed that maximum percent of general public are reacting positively to fundraising campaigns and initiatives.

Analysis based on the theme of sentiments :

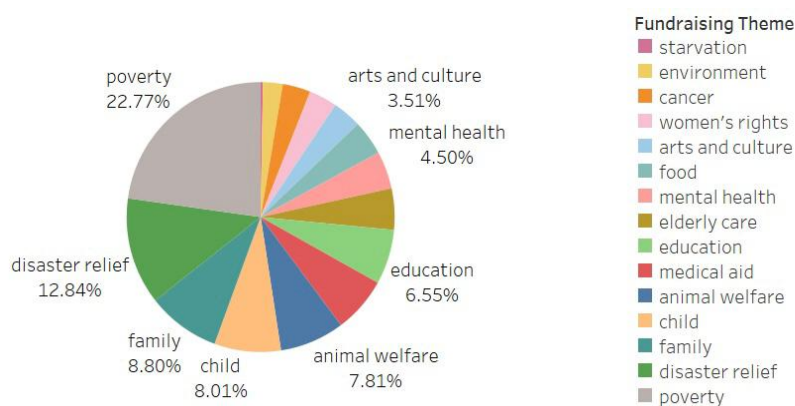
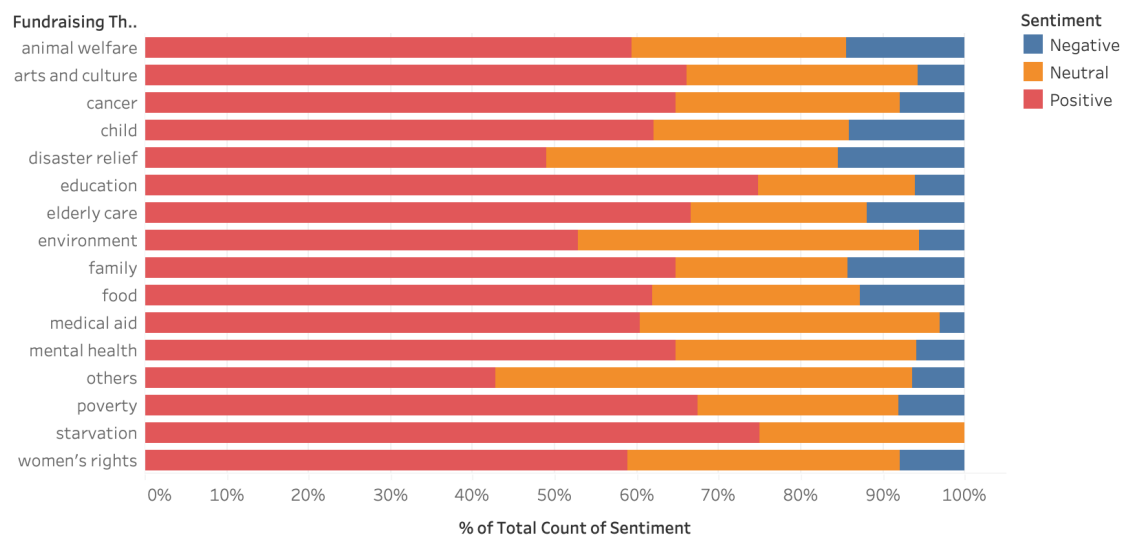


Figure-10 : Figure showing the distribution of sentiments based on fundraising themes

1. The above chart shows significant trends with respect to fundraising themes. From our analysis we have obtained significant insights into fifteen fundraising themes.
2. We see that poverty accounting for 22.77% is the most popular theme, followed by disaster relief consisting up of 12.84% and family child welfare accounting for 8.80% and 8.01% respectively .
3. Thus we can understand that the top five social topics are the ones where public is more concerned about.

Sentiment Distribution of fundraising themes:

Sentiment Distribution of Fundraising Themes



% of Total Count of Sentiment for each Fundraising Theme. Color shows details about Sentiment. The data is filtered on Tweet Timestamp, which ranges from 29/01/2019 to 30/01/2024. The view is filtered on Sentiment, which keeps Negative, Neutral and Positive.

Figure-11: The above graph represents the sentiment distribution of various fundraising themes

1. The above graph represents sentiment distribution across various fundraising themes
2. Animal welfare , disaster relief and family welfare received maximum negative comments accounting for 14.41%, 15.46% and 14.29% respectively
3. Starvation, Education and Poverty received maximum positive comments consisting up of 75%, 74.75% and 67.44%

TABLEAU DASHBOARD:

1. DEMOGRAPHIC DASHBOARD:

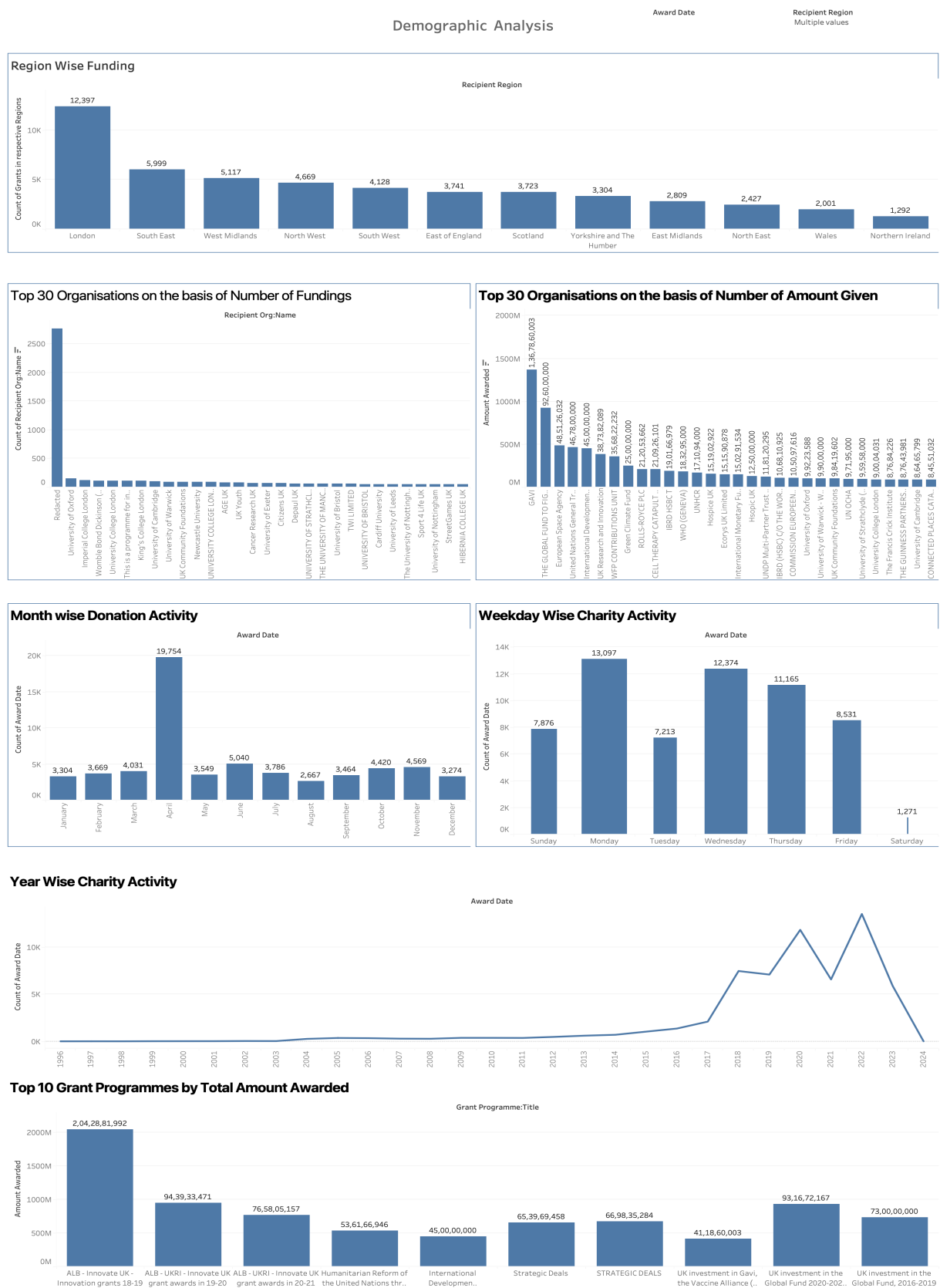


Figure-12: The above figure represents demographic tableau dashboard.

1. We created this dashboard to visualise demographic data automatically by using the TabPy library of python.
2. From this demographic dashboard we can find that we can select different time periods and recipient regions and all the graphs will change.

2. SENTIMENT DASHBOARD:

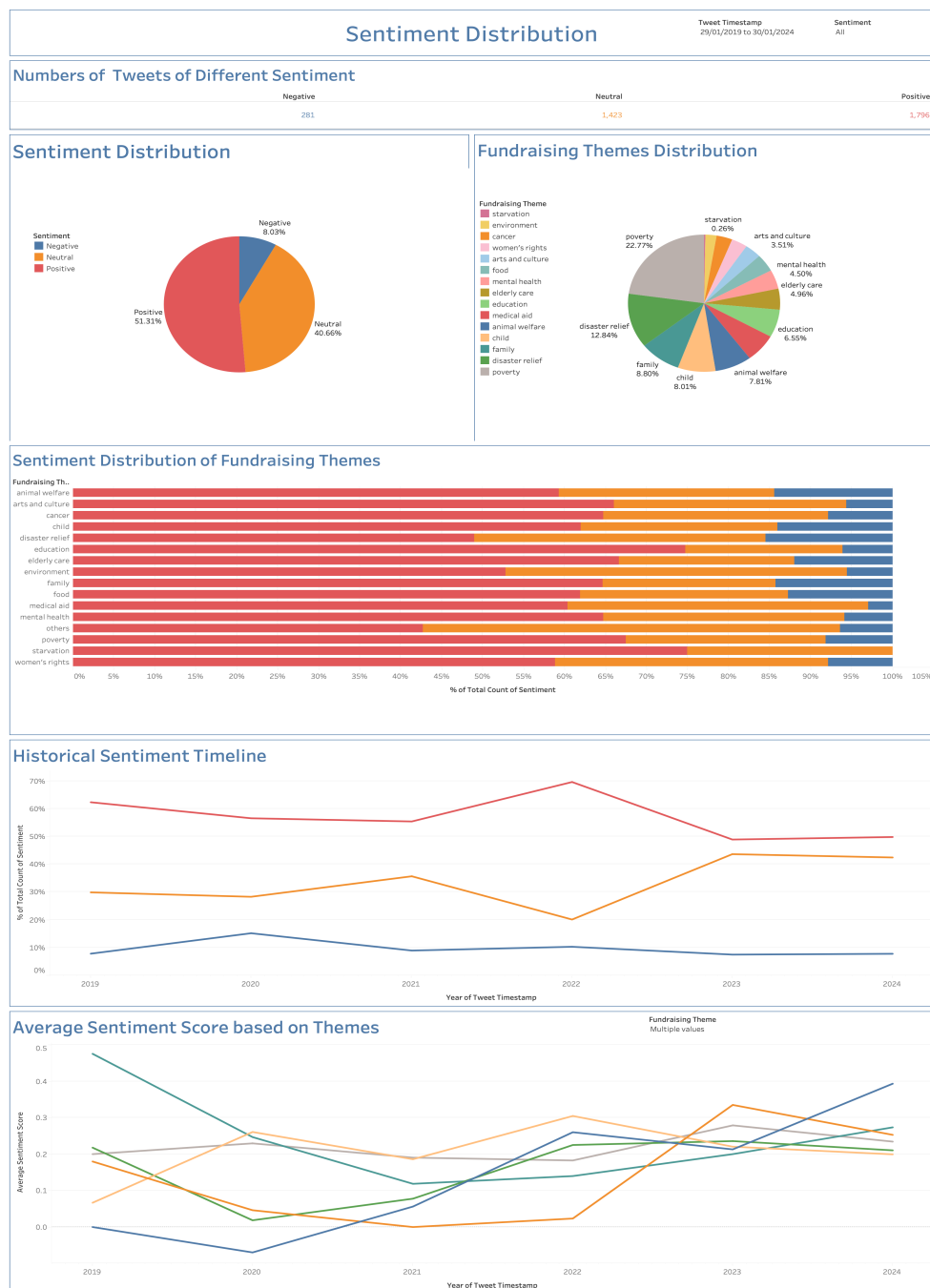


Figure-13: The above figure represents sentiment tableau dashboard.

1. We created this dashboard to visualise sentiment data automatically by using the TabPy library of python.
2. In the sentiment dashboard we can select specific time periods or specific sentiments that we want to further investigate.
3. Apart from the basic sentiment distribution this dashboard also contain graph related to year wise analysis of sentiments from 2019 to 2024.

Suggestion for the Client:

1. With London being a high-engagement area with 24% of Donations, It is necessary to continue fundraising efforts there but also consider a strategic expansion into low engagement regions like Northern Ireland or Wales. Focused campaigns in these regions can help in diversifying and expanding the donor base.
2. Partnering with top funding organizations like Global Fund, Gavi, WFP contribution programme, etc will help in accessing their networks for a widespread reach and to facilitate partnered events that can lead to more meaningful fundraising opportunities.
3. We found that the number of donations seems to be very high in April around 32% as it marks the end of the financial year and charity donors get tax exemptions, hence intensifying marketing efforts during that time can help us in getting more funds.
4. We analysed the demographic data to identify the sector with highest income potential, we found that private sector workers and business owners have higher income and they make the maximum donations around 73.7%. Due to market dependent competitive salaries and performance based incentives private sector employees have higher income compared to others (Fernandes, Ferreira, & Winters, 2017). So, we need to focus on workplace campaigns in the private sector where there are higher earners and involve them in corporate social responsibility (CSR) programs, by which we can improve employee engagement and increase donations.
5. Through a combination of word cloud and sentiment analysis we found that the positive sentiments expressed around fundraising themes like 'Poverty', 'Disaster Relief' and 'Family Welfare' were maximum with 22.77%, 12.84% and 8.80% respectively. Developing campaigns related to these themes or the intersection of the three areas, such as help for children in poverty stricken areas affected by disasters can help us find more corporate sponsors whose CSR goals align with these themes for broader reach

and increased funding. Partnering with organizations that specialize in these fields we can maximize resources and its impact on the public.

6. Themes like 'disaster relief' and 'animal welfare' have maximum negative opinion in social media, so by reevaluating campaign strategies, enhancing social media presence and conducting public outreach initiatives to explain the seriousness and impact of these issues can help our goals align with donor expectations, improve sentiment of the public and donation activity in these areas.
7. Analysing the census data we found that middle aged individuals who belong to the age group of 37 to 47 have higher income potential. Hence aligning fundraising strategies specific to the individuals of this age group can help us secure larger donations. Giving hand written letters, recognition titles or gift merchandise to donors for their generous donations can make a better impression and reduce donor churn (Merchant, Ford, & Sargeant, 2010).
8. Charities can use automated Tableau dashboards to enhance data driven decision making and reduce manual operation (Hoelscher & Mortimer, 2018). By visualizing graphs like donation trends, donor demographics and fund allocation these dashboards can give us a comprehensive analysis of organizational performance. This will help managers to identify areas needing attention and tailor fundraising strategies to increase their impact.

Effective Team Work:

Initial Planning, Scheduling and Progress Tracking:

The success and quality of a group project largely depends on the quality of the initial meeting and using a structured technique that involves problem identification, solution generation, and decision-making (Asmus and James, 2005). In the first week of our project, we assigned each member with suitable roles based on their strengths and experience (Appendix-V). Meanwhile, we set a schedule for our offline meetings on every Tuesday and Sunday, in which we report on project progress, discuss, and resolve current issues, and meticulously divide tasks for each member to complete before the next meeting. The entire project spanned for Fifteen weeks. We made Gantt chart (Appendix-IV) using Microsoft Project to track the progress of our project. Initially we had a comprehensive online meeting with our supervisor and our client to thoroughly understand the project's objectives and the client's requirements. During each group session we thoroughly assessed the project's ongoing progress. Being the group leader,

Rudrasish ensured that the project align to the timeline completely, avoiding any delays. This deliberate approach ensured that each phase of the project received the attention it deserved and was executed properly.

Team Communication and Decision Making:

Communication is one of the crucial factors in the project management process, effective communication practices in project lifecycle ensures the overall success of the project (Muszyńska, 2016). Other than regular offline meetings, we conducted online meetings through Google Meet when necessary. We mainly used Gmail, WhatsApp and Google chat to share information. All project documents were managed using a google drive folder to easy track the work of each team member. Access control protocols were setup so that modifications to files on drive was approved by the whole team, thereby guaranteeing that tasks met the expectations of all team members. The ambiance within our team was consistently positive, fostering an environment for proactive support and mutual encouragement. Decision making is a collaborative process where each member has the opportunity to give their opinion (Nikoi and Boateng, 2013).

Data Collection:

The project was divided into three phases data collection, data analysis and report writing. During the data collection phase we gathered data about fundraising campaigns, but faced difficulties due to data privacy rules and unavailability of reliable data resources related to fundraising campaigns. To solve this issue related to data collection we had interactions with our unit director, co-director, supervisor and client. The suggestions that we got from them helped us in collecting data by scraping Twitter(X), Instagram and Facebook. Eventually, Rudrasish successfully collected relevant data related to fundraising organizations from sources like UK Data Service, 360 Giving (GrantNav), and UK Government website, while Yiheng used Octoparse to scrape user comments and posts on fundraising activities from Twitter(X).

Programming and Analysis:

In the programming phase, Rudrasish entirely took charge of writing the program, applying necessary machine learning methods including regression analysis, random forest, time series

forecasting and clustering to analyse fundraising organizations. Firstly ROI and demographic analysis was done on the data collected from 360 Giving (GrantNav) to understand the correlation between various factors and identify the top regions, donors, organisation and recipients involved in fundraising activities. Rudrasish and Yiheng worked collaboratively performing sentiment analysis on Twitter(X) posts, eventually automating the process of visualisation in Tableau by using the TabPy library. Rudrasish performed donor analysis on the census data to recognise possible donors for the fundraising campaign and a tableau dashboard was created to automatically visualise the outcome. Yiheng and Srishti were responsible for verifying the programming results, check the model accuracy and to ensure the scripts ran correctly.

Report Writing and Collaboration:

The report writing was primarily handled by Yiheng and Srishti with the analytical problem part written by Rudrasish and Yiheng. The Analytics Solution, Technical Development, and Effective Teamwork sections were written by Yiheng and Srishti. The entire report writing process was overseen, reviewed and modified by Rudrasish carefully who offered guidance and support in each and every step.

References:

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Fernandes, A., Ferreira, P., & Winters, L., 2017. The Effect of Competition on Executive Compensation and Incentives. *The Journal of Human Resources*, 53, pp. 783 - 824. <https://doi.org/10.3368/jhr.53.3.0215-6963R1> (Accessed Date: 2 April 2024).

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APPENDIX

Appendix-1

CODE FILES:

1. **Demographic Analysis:**
<https://colab.research.google.com/drive/1aa0issjd1x74zEqBhs0p9Iyi3KryDx-G?usp=sharing#scrollTo=sjvLzFY2hvFL>
2. **Effectiveness analysis:**
<https://colab.research.google.com/drive/1vRAD955VAORwTUIIn5yeAkzNdJQ0ikbtY>
3. **Categorical Analysis:**
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4. **Forecasting:**
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5. **Donor Prediction:**
<https://colab.research.google.com/drive/1ziUPmxQX6ax4fP3p5W8eN7Lezd1pciwt?usp=sharing>
6. **Sentiment Analysis:**
https://colab.research.google.com/drive/1XUdy_WxVeOevYmM6x00f6ZMbX_bf_vhFk?usp=sharing
7. **Cause Analysis:**
<https://colab.research.google.com/drive/1ew76ui7eHD5ApPejD1ffhfV1mlx8vDrD>

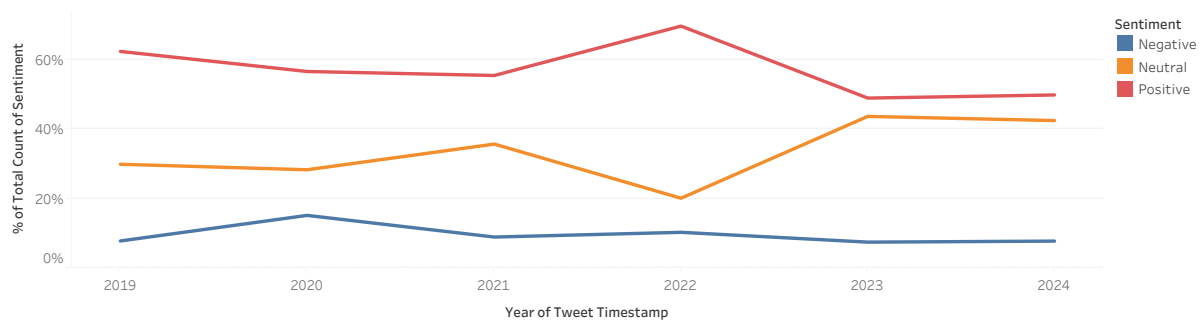
Appendix-2

Tableau Files:

<https://drive.google.com/drive/folders/1bGzEgGz9pijvP3pxuuqHIeXn24z3SeLx?usp=sharing>

Appendix-3

Historical Sentiment Timeline



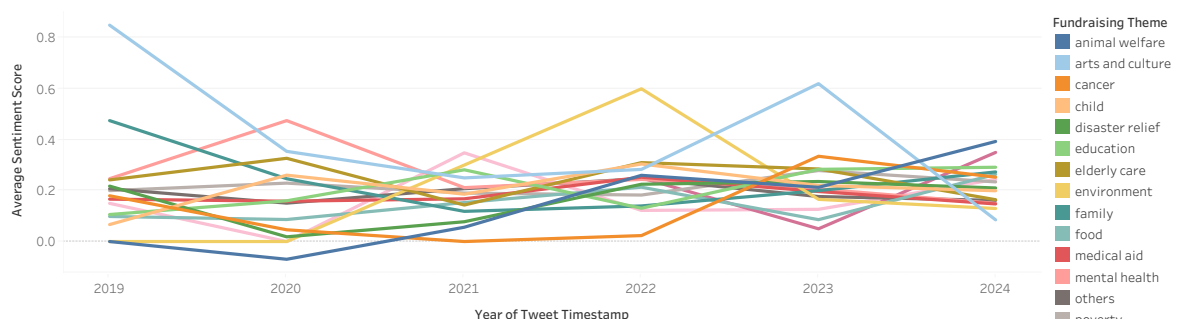
The trend of % of Total Count of Sentiment for Tweet Timestamp Year. Color shows details about Sentiment. The data is filtered on Tweet Timestamp, which ranges from 29/01/2019 to 30/01/2024. The view is filtered on Sentiment, which keeps Negative, Neutral and Positive.

Figure-14: This graph represents average sentiment scores across the years.

1. The above graph represents the trends of three different sentiments (positive, negative and neutral) expressed in tweets from 2019 to 2024.
2. The line for positive sentiment starts at around 60% in 2019 then there is a gradual decline until 2021 and from 2022 onwards, there is a slight noticeable increase till the initial phase of 2024.
3. This line for negative sentiment shows a decreasing trend from 2019 to 2021 further it stabilizes slightly above 10% before beginning a slight increase around 2022 till 2024.

Appendix-4

Average Sentiment Score based on Themes



The trend of Average Sentiment Score for Tweet Timestamp Year. Color shows details about Fundraising Theme. The data is filtered on Sentiment and Tweet Timestamp. The Sentiment filter excludes Null. The Tweet Timestamp filter ranges from 29/01/2019 to 30/01/2024. The view is filtered on Fundraising Theme, which keeps multiple members.

Figure-15: The above graph represents average sentiment scores based on themes.

1. It is vividly shown from the graph that the sentiment score of the fundraising themes for each year fluctuate significantly, this may indicate that public's attitudes towards specific social topics may be affected by certain event.
2. For those fundraising themes with increasing sentiment scores, charities should focus on allocating more resources into them.

Appendix-5

	A	B	C	D	E	F	G	H	I
1	DURO INSIGHT TEAM MEETING					TEAM MEMBERS ATTENDANCE SCHEDULE			
2	DATE	REASON	LOCATION	DAY		Rudrasish Mishra	Yiheng Guo	Srishti Anand	Debmalya Das
3	27th February	Team Meeting with Professor due to Data Collection Issue	Professor Akinyo Ola's Office	Tuesday			Reason : Travelling		
4	29th February	To discuss further steps after data collection	Online	Thursday					
5	3rd March	To work on the project plan due to approaching submission date	St Thomas Street Study Space	Sunday					Arrived Late due to Travelling
6	10th March	To work on the Project Initial Stage coding and report Writing	St Thomas Street Study Space	Sunday					Arrived Late
7	12th March	To work on academic resource collection and extraction...	St Thomas Street Study Space	Tuesday					
8	17th March	To discuss about the coding part and future work section of the report.	St Thomas Street Study Space	Sunday					
9	19th March	To discuss about project progress and improving data collection process	St Thomas Street Study Space	Tuesday					
10	EASTER BREAK (No Meetings till 31st March)								
11									
12									
13									
14	3rd April	To discuss about the Grading and further improvement Scope	Professor Akinyo Ola's Office	Wednesday					Not Informed
15	7th April	To Work on Report Writing and discuss on Data Analysis	St Thomas Street Study Space	Sunday					
16	9th April	To Work on Report Writing and discuss on Sentiment Analysis	St Thomas Street Study Space	Tuesday					
17	14th April	To Work on Report Writing	St Thomas Street Study Space	Sunday		MEETING CANCELLED (HAD A REPORT WRITING SESSION)			
18	16th April	To Work on Report Writing and discuss on Sentiment Analysis	St Thomas Street Study Space	Tuesday					
19	18th April	To discuss about the Grading and further improvement Scope	Professor Fatima's Office	Thursday					
20	20th April	Unscheduled Meeting to Work on Report Writing	St Thomas Street Study Space	Saturday					Not Informed
21	21st April	To Work on Report Writing and ROI & Sentiment Analysis	St Thomas Street Study Space	Sunday					
22	23rd April	To Work on Report Writing	St Thomas Street Study Space	Tuesday					
23	26rd April	Client Meeting	Online	Friday					
24	26th April	Unscheduled meeting to Work on Report Writing and Video Presentation	St Thomas Street Study Space	Friday					Not Informed
25	27th April	Unscheduled Meeting to Work on Report Writing and Video Presentation	St Thomas Street Study Space	Saturday					Not Informed
26	28th April	To Work on Report Writing and Video Presentation	St Thomas Street Study Space	Sunday					
27	TOTAL NO OF MEETINGS ATTENDED PROPERLY					0 18 18 0	0 17 18 1	0 18 18 0	4 6 13 4
28	REMARKS ON THE BASIS OF CONTRIBUTION, PARTICIPATION AND INVOLVEMENT					Perfect leader!Team player! A man worthy to be trusted and work with	Extraordinary contribution in data collection, Analysis and Report writing. Attended all the meetings with utmost dedication. Is a really good team player	Exhibited outstanding leadership skills in heading the report writing part . Attended all the meetings with full sincerity. Is a really good team player	Have scope for improvement. Need to show dedication and involvement in Teamwork.
29	OVERALL REMARKS ON THE BASIS OF CONTRIBUTION, PARTICIPATION AND INVOLVEMENT					EXCELLENT	EXCELLENT	EXCELLENT	SATISFACTORY
30	REMARKS MADE BY					Yiheng Guo	Srishti Anand and Rudrasish Mishra	Yiheng Guo and Rudrasish Mishra	Yiheng Guo, Srishti Anand and Rudrasish Mishra

Figure-16: Team Meeting Schedule and remarks for each member.

Appendx-6

TABLEAU DASHBOARD FOR DEMOGRAPHIC ANALYSIS:

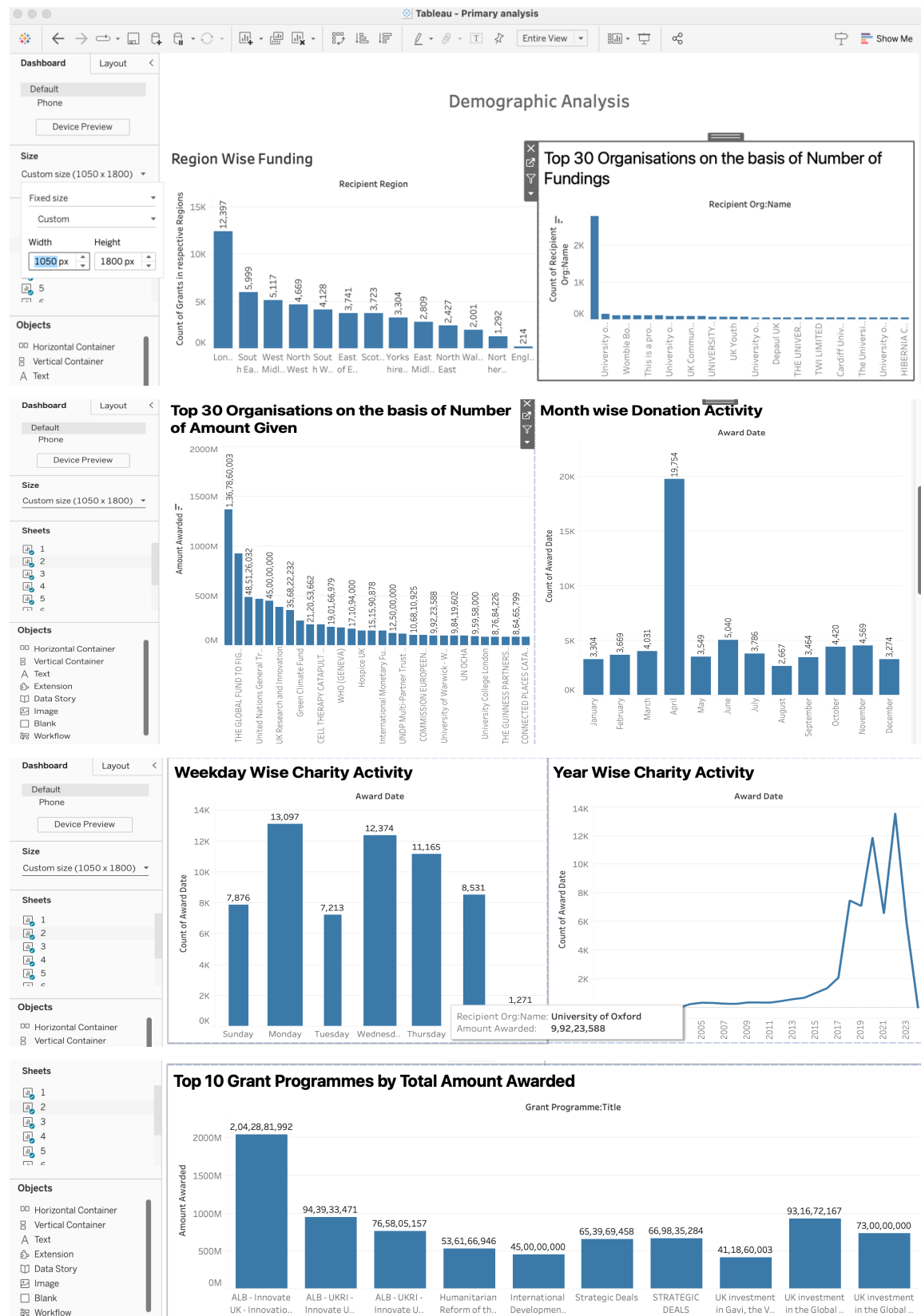


Figure-17: The above figure represents the entire demographic tableau dashboard.

Appendix-7

TABLEAU DASHBOARD FOR SENTIMENT ANALYSIS:

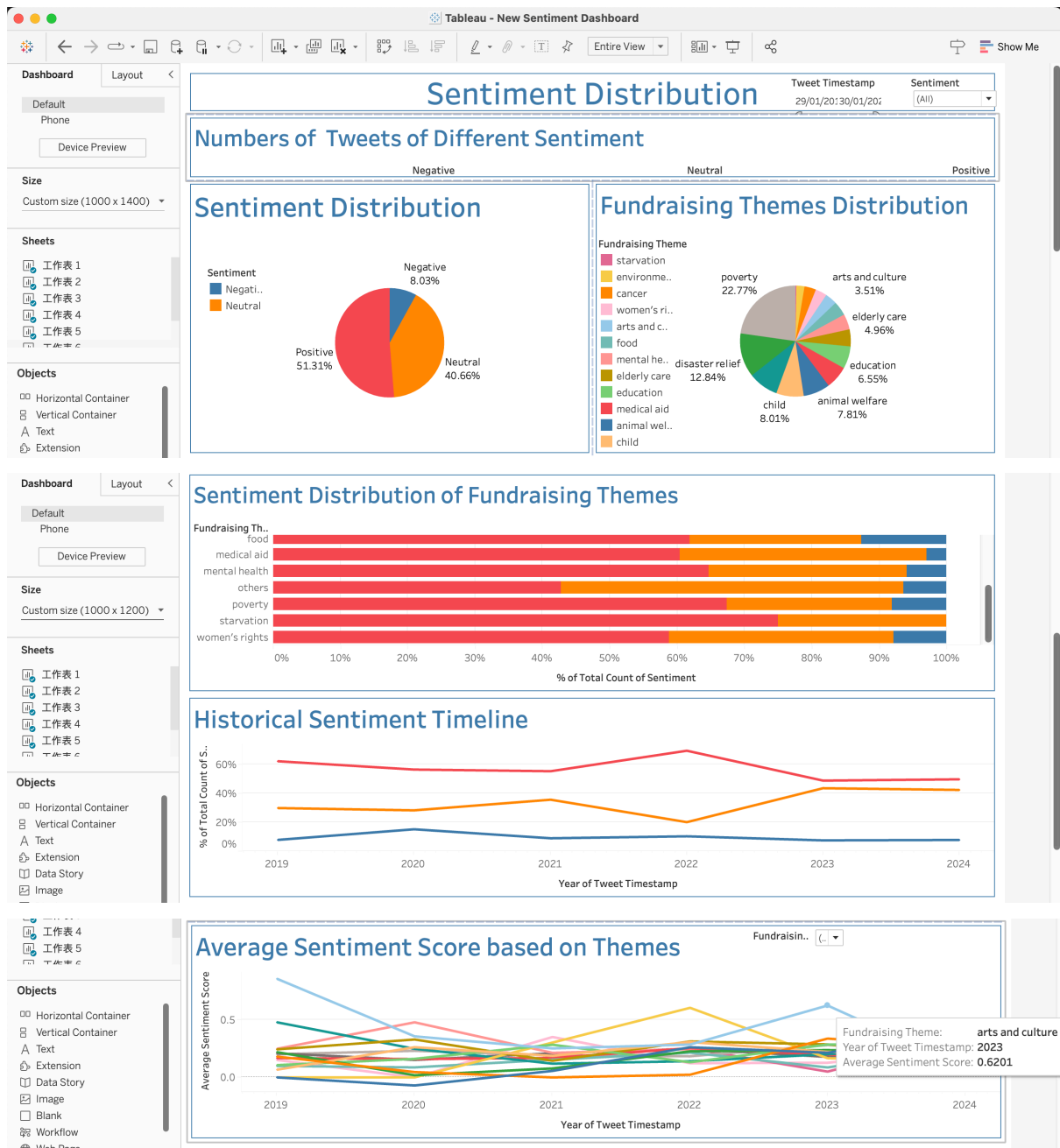


Figure-18: The above figure represents entire sentiment tableau dashboard.

Appendix-8

Causal Loop Diagram:

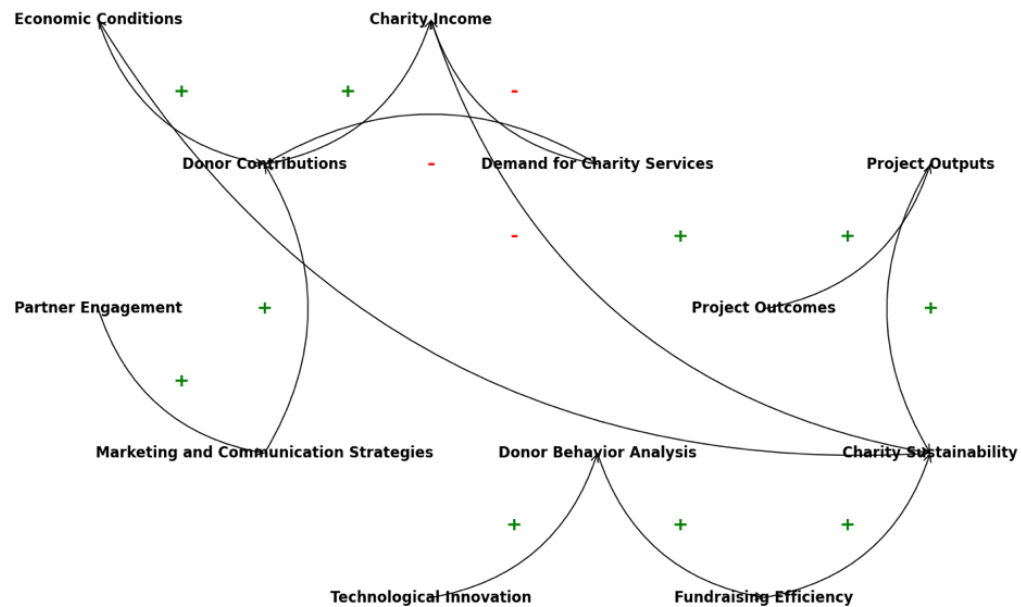


Figure-19 : Causal Loop Diagram for Fundraising Campaigns

1. This casual diagram is a simplified representation of how different factors interact in the in a charity organization.
2. The circles (nodes) and labels represent various elements like Economic Conditions, Charity Income, Fundraising Efficiency, etc. and arrows show the direction of influence,
3. A plus sign (+) means that if one factor increases, the other also increase, while a minus sign (-) indicates that an increase in one factor leads to a decrease in the other and vice versa. This diagram will help us in visualising the complex relationships between different aspects of a charity operations.

Appenxdix-09

FEEDBACK FROM THE CLIENT:

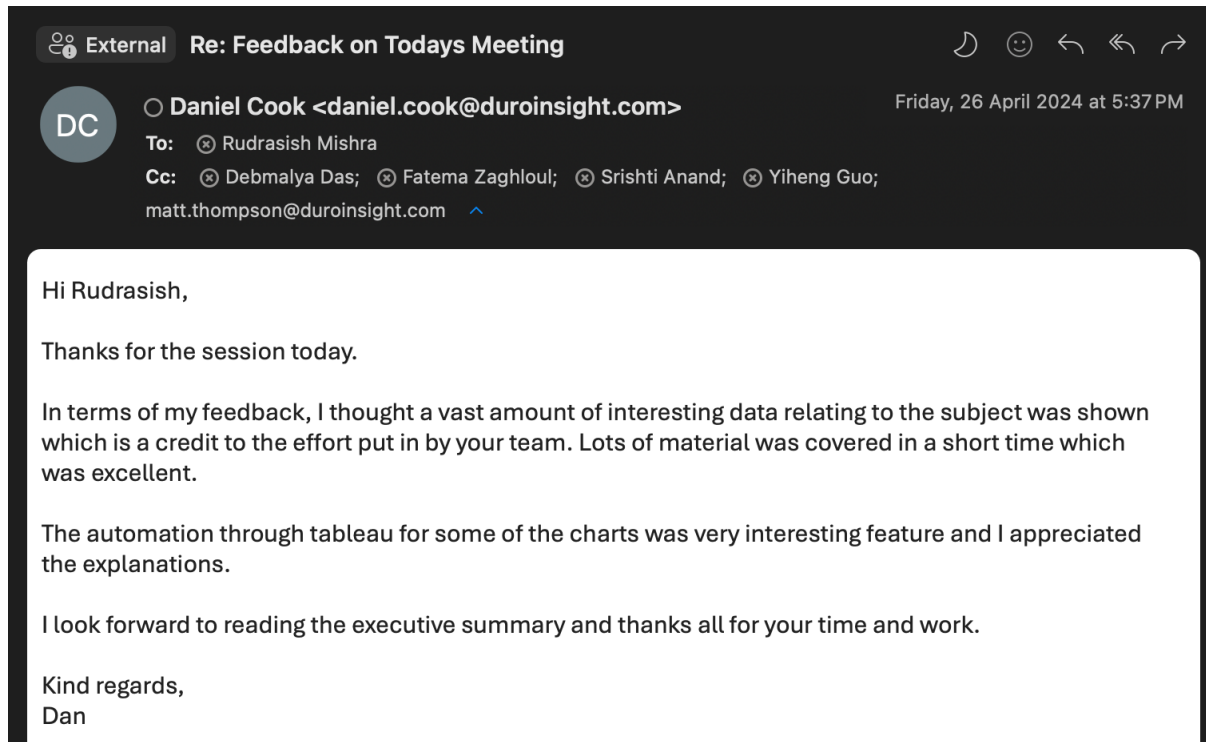


Figure-20: Feedback from the client after presentation.

Appenxdix-10

Contribution Statement :

Student number	Student name	Areas of Contribution	Rating
2403296	Rudrasish Mishra	Data Collection, Report Writing, Project Management, Data Analytics and Project Planning	Excellent
2382612	Yiheng Guo	Data Collection, Data Analysis, Report Writing and Project Planning	Excellent
2393081	Srishti Anand	Report Writing, Presentation and Project Planning	Excellent
2357265	Debmalya Das		Satisfactory

Table-1: The above table contains the different contributions made by each and every team member in different phases of the project.

Appendix-11

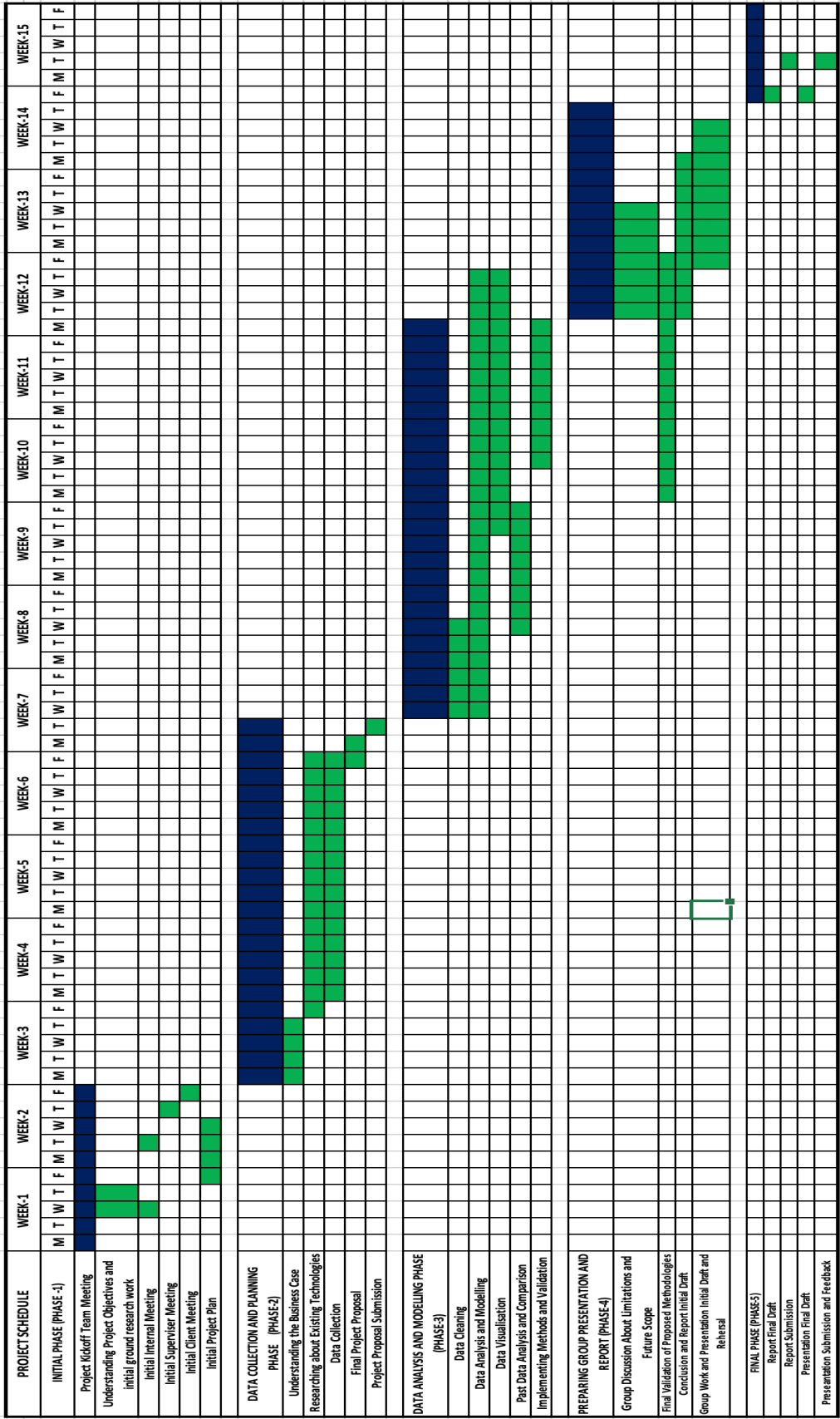


Figure-21: Detailed Timeline of the project (Project Schedule)